

The BGG correspondence: strategy of the classical proof

1 Statement

Let V be an $(n+1)$ -dimensional vector space over a field $\mathbf{k}$, $W = V^*$, and

$$S = \text{Sym}(V), \quad E = \Lambda(W),$$

so that $\mathbb{P}^n = \mathbb{P}(W)$ has homogeneous coordinate ring S . Grade S by $\deg V = 1$ and E by $\deg W = -1$. The algebra E is a graded Frobenius algebra with $\omega_E \cong \text{Hom}_{\mathbf{k}}(E, \mathbf{k}) \cong E(-n-1)$ up to shift convention, so finitely generated free E -modules are both projective and injective.

Theorem 1 (Bernstein–Gelfand–Gelfand, 1978). *There is an equivalence of triangulated categories*

$$\text{D}^b(\text{coh}\mathbb{P}^n) \simeq \underline{\text{grmod}}E \simeq \text{D}^b(\text{grmod}E)/\text{perf}(E),$$

where $\underline{\text{grmod}}E$ is the stable category of finitely generated graded E -modules.

The proof has three ingredients: Koszul duality between S and E , a comparison of the two Serre/Verdier quotients, and (for explicit inverses) the Tate resolution and Beilinson’s resolution of the diagonal.

2 Step 1: Koszul duality

For a graded S -module M and a graded E -module P define the linear complexes

$$\mathbf{R}(M) : \cdots \rightarrow \text{Hom}_{\mathbf{k}}(E, M_d) \xrightarrow{\sum e_i \otimes x_i} \text{Hom}_{\mathbf{k}}(E, M_{d+1}) \rightarrow \cdots \quad (\text{free } E\text{-modules}),$$

$$\mathbf{L}(P) : \cdots \rightarrow S \otimes_{\mathbf{k}} P_d \xrightarrow{\sum x_i \otimes e_i} S \otimes_{\mathbf{k}} P_{d+1} \rightarrow \cdots \quad (\text{free } S\text{-modules}),$$

where x_i, e_i are dual bases of V, W . The module structure on one side becomes the differential on the other.

Theorem 2 (BGG, Koszul duality). *$\mathbf{R}$ and $\mathbf{L}$ induce mutually inverse equivalences*

$$\text{D}^b(\text{grmod}S) \simeq \text{D}^b(\text{grmod}E).$$

Idea. The Koszul complex $S \otimes \Lambda(V)$ resolves $\mathbf{k}$ over S , and dually over E . Hence $\mathbf{R}(S(d))$ is, up to shift, the residue field $\mathbf{k}$ over E , while $\mathbf{R}(\mathbf{k})$ is a free module (ω_E up to shift). Since both categories are generated by these objects, and Hom spaces between generators match (Koszul property), $\mathbf{R}$ is fully faithful and essentially surjective. (Shifts and twists depend on conventions.)

3 Step 2: Passing to the quotients

Recall $\mathrm{coh}\mathbb{P}^n \simeq \mathrm{grmod}S/\mathrm{grmod}_{\mathfrak{h}}S$, the quotient by finite-length (torsion) modules (Serre), so

$$\mathrm{D}^b(\mathrm{coh}\mathbb{P}^n) \simeq \mathrm{D}^b(\mathrm{grmod}S)/\mathrm{D}_{\mathfrak{h}}^b(\mathrm{grmod}S).$$

Finite-length modules are built from copies of $\mathbf{k}(d)$, and $\mathbf{R}(\mathbf{k})$ is free over E . Therefore $\mathbf{R}$ carries $\mathrm{D}_{\mathfrak{h}}^b(\mathrm{grmod}S)$ onto the thick subcategory $\mathrm{perf}(E)$ of perfect complexes, and Koszul duality descends to

$$\mathrm{D}^b(\mathrm{coh}\mathbb{P}^n) \simeq \mathrm{D}^b(\mathrm{grmod}E)/\mathrm{perf}(E).$$

Finally, since E is Frobenius, $\mathrm{grmod}E$ is a Frobenius category, and the natural functor $\underline{\mathrm{grmod}}E \rightarrow \mathrm{D}^b(\mathrm{grmod}E)/\mathrm{perf}(E)$ is an equivalence (Rickard, Buchweitz). This proves the theorem.

4 Step 3: Explicit form via Tate resolutions

For $\mathcal{F} \in \mathrm{coh}\mathbb{P}^n$ the equivalence can be made explicit. Let $M = \bigoplus_d H^0(\mathcal{F}(d))$ be its module of twisted sections. Applying $\mathbf{R}$ to high truncations $M_{\geq r}$ gives an exact complex of free E -modules in high degrees; extending to the left by a free resolution yields the *Tate resolution* $\mathbf{T}(\mathcal{F})$, an acyclic complex of free E -modules with terms

$$\mathbf{T}(\mathcal{F})^e = \bigoplus_i \omega_E(e-i) \otimes H^i(\mathcal{F}(e-i)).$$

Thus $\mathbf{T}(\mathcal{F})$ packages all cohomology groups of all twists of $\mathcal{F}$. Syzygy modules of $\mathbf{T}(\mathcal{F})$ are the objects of $\underline{\mathrm{grmod}}E$ corresponding to $\mathcal{F}$ (up to shift), and $\mathbf{T}$ is the functor realizing the equivalence of the theorem on objects.

5 Step 4: Beilinson's resolution of the diagonal

On $\mathbb{P}^n \times \mathbb{P}^n$ there is the exact sequence

$$0 \rightarrow \mathcal{O}(-n) \boxtimes \Omega^n(n) \rightarrow \cdots \rightarrow \mathcal{O}(-1) \boxtimes \Omega^1(1) \rightarrow \mathcal{O} \boxtimes \mathcal{O} \rightarrow \mathcal{O}_{\Delta} \rightarrow 0.$$

It yields Beilinson's monad: every $\mathcal{F}$ is the cohomology of a complex with terms $\bigoplus_j H^i(\mathcal{F}(-j)) \otimes \Omega^j(j)$, whose differentials are read off from $\mathbf{T}(\mathcal{F})$. In particular $\mathcal{O}(-n), \dots, \mathcal{O}$ is a full exceptional collection, so $\mathrm{D}^b(\mathrm{coh}\mathbb{P}^n)$ is generated by finitely many explicit objects. This is the geometric counterpart of the algebraic generation used in Step 1, and gives an alternative route to full faithfulness and essential surjectivity: check on the generators $\Omega^j(j)$, which correspond to the simple modules $\mathbf{k}$ (suitably shifted) over E , then extend by dévissage.

6 Summary of the strategy

1. **Koszul duality:** $\mathrm{D}^b(\mathrm{grmod}S) \simeq \mathrm{D}^b(\mathrm{grmod}E)$, proved on generators $(S(d) \leftrightarrow \mathbf{k})$ and extended by generation.

2. **Quotients:** torsion S -modules correspond to perfect E -complexes, so Serre localization on the S -side matches Verdier localization on the E -side.
3. **Frobenius property:** $D^b(\text{grmod} E)/\text{perf}(E) \simeq \underline{\text{grmod}} E$.
4. **Explicit inverses:** Tate resolutions and Beilinson monads encode sheaf cohomology and recover $\mathcal{F}$ from the E -module.

Remark 3. Gradings, shifts, and which of V , V^* appears vary between sources, so exact twists above should be checked against a reference. A readable account is Eisenbud–Fløystad–Schreyer, *Sheaf cohomology and free resolutions over exterior algebras*, Trans. AMS 355 (2003).